

Numerical Project: Control of Acoustic Wave by Damping on a Surface

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Abstract

This report details the theoretical analysis and numerical results for determining the optimal acoustic damping coefficient $\alpha(\omega)$ that models a porous wall as an equivalent impedance boundary condition. The analysis covers the dependence of α on various source spectra and compares the efficiency of different porous materials (Melamine foam, Glass wool, Rock wool, and Polyurethane foam).

Question 1

In the presented theorem, the source g depends only on the spatial variable y . We now suppose that g also depends on the frequency ω .

The theorem remains entirely similar, and no modification is necessary other than specifying the dependence on ω in the boundary condition, i.e., using $g(y, \omega)$ instead of $g(y)$. The Fourier coefficients g_k will thus be functions of ω as well, $g_k(\omega)$.

Question 2

Below are three examples of physical situations where absorbing acoustic energy is crucial, along with the corresponding models for the source $g(y, \omega)$.

Case 1 – Noise Barrier along a Highway (Source g_1)

Motivation. Noise barriers protect residential areas from road noise. The primary sound sources are engines and tire rolling, producing a spectrum dominated by low frequencies with a decay across the frequency range.

Model. The noise is assumed to be uniform along the wall (negligible spatial dependence on y) and to have an amplitude that decays exponentially with frequency, concentrating energy at low frequencies:

$$g_1(y, \omega) = A_1 e^{\frac{-\omega}{\omega_c}},$$

where ω_c characterizes the average frequency of the noise spectrum.

Case 2 – Industrial Turbine or Aircraft Engine (Source g_2)

Motivation. Turbines generate powerful sounds that are both broadband and harmonic. The absorption strategy must reduce both the broadband continuum and the discrete harmonic tones.

Model. The source (approximated as uniform along the wall) is represented as the superposition of a decaying broadband noise and harmonic components:

$$g_2(y, \omega) = \left(C \frac{1}{1 + (\omega/\omega_c)^p} \times \sum_{m=1}^M A_m \exp \left(- \frac{(\omega - \omega_m)^2}{2\sigma_{\omega, m}^2} \right) \right).$$

The first term represents the broadband background noise, and the peaks $\omega_m = m\omega_0$ model the harmonics of a fundamental frequency ω_0 .

Case 3 – Ventilation System (Source g_3)

Motivation. Ducts and fans produce monotone low frequencies that must be absorbed for internal acoustic comfort.

Model. The source is localized around a point y_0 (Gaussian in y) and centered on a dominant frequency (Gamma distribution in ω , maximizing entropy knowing the average and the variance of the energy distribution over ω):

$$g_3(y, \omega) = A_3 \omega^{k-1} \exp \left(- \frac{(y - y_0)^2}{2\sigma_y^2} - \frac{\omega}{\theta} \right),$$

where θ , k , and σ_y control the spectral width and the spatial localization of the source.

Question 3 – Implementation and Results for One Material

We consider a fixed porous material, a melamine foam characterised by the following physical parameters:

$$\phi = 0.99, \quad \sigma = 14000.0 \text{ N}\cdot\text{s}\cdot\text{m}^{-4}, \quad \alpha_h = 1.02.$$

The frequency range is $\omega \in [2\pi \times 100, 2\pi \times 5000] \text{ rad}\cdot\text{s}^{-1}$. The minimization constants are set to $A = 1$ and $B = 1$.

Method. The objective is to find the complex boundary coefficient $\alpha(\omega)$ that minimizes the functional:

$$J(\alpha) = A\|u_0 - u_2\|_{L^2(\Omega_{\text{air}})}^2 + B\|\nabla(u_0 - u_2)\|_{L^2(\Omega_{\text{air}})}^2,$$

which measures the discrepancy between the full coupled problem (air + porous wall, u_0) and the simplified air-only problem with impedance boundary condition $\eta_0 \nabla u_2 \cdot \nu + \alpha u_2 = 0$ (u_2). In the frequency domain, this minimization is performed by minimizing the error function $e(\alpha) = \sum_k e_k(\alpha)$.

The optimal α is found to be related to the surface impedance $Z_s(\omega)$ of the porous medium (e.g., via the Johnson–Champoux–Allard model for $\rho(\omega)$ and $K(\omega)$):

$$\alpha(\omega) \approx \frac{\rho_0 c_0}{Z_s(\omega)}, \quad Z_s(\omega) = \sqrt{\rho(\omega) K(\omega)}.$$

The $\alpha(\omega)$ values are thus calculated using this optimal closed-form expression, which is derived from the minimization problem.

Numerical implementation. The code implements the JCA model to determine the frequency-dependent effective properties ($\rho(\omega), K(\omega)$) based on the Melamine foam parameters. The optimal coefficient $\alpha(\omega)$ is then calculated for a range of frequencies ω . For comparison, the resulting α is typically weighted by the normalized source spectrum of each $g_i(y, \omega)$.

Results. Figure 1 shows the real and imaginary parts of $\alpha(\omega)$ for the three sources.

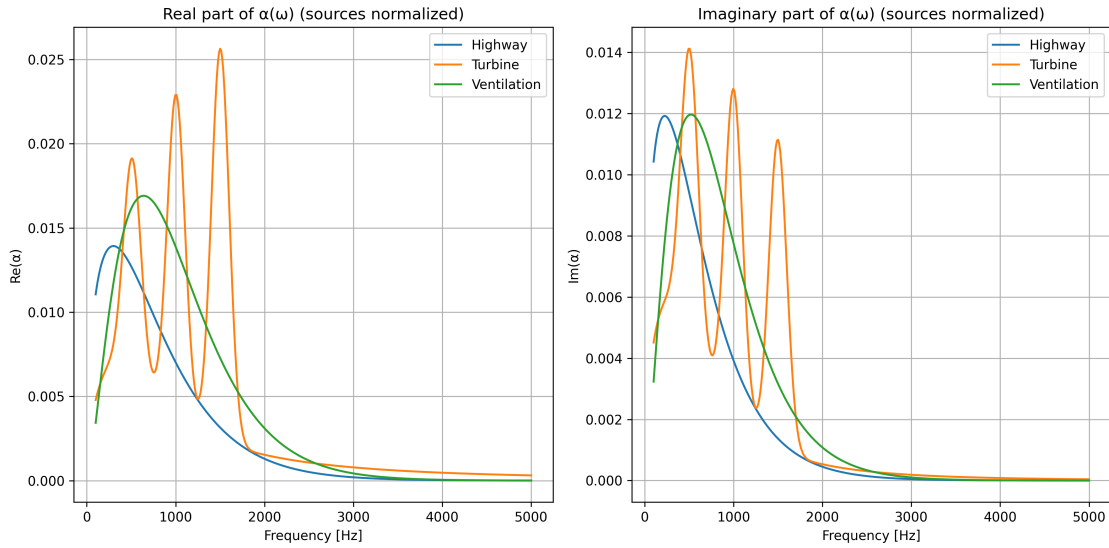


Figure 1: Real and imaginary parts of $\alpha(\omega)$ for melamine foam and the three types of acoustic sources.

Discussion. The real part $\Re(\alpha)$ represents the dissipative component, responsible for the true absorption of acoustic energy. The imaginary part $\Im(\alpha)$ is the reactive component, linked to the phase shift between pressure and velocity at the surface (energy temporarily stored).

- Highway Noise (g_1): Dominated by low frequencies, it generally requires high $\Re(\alpha)$ at small ω , corresponding to strong viscous damping in that range.
- Turbine Noise (g_2): With a broadband spectrum and harmonics, it yields a more oscillatory $\Re(\alpha)$ and $\Im(\alpha)$ across the range, reflecting the need for optimal absorption at discrete frequencies.
- Ventilation Noise (g_3): Narrow and low-frequency, the response presents a significant imaginary component at low ω , characteristic of a reactive boundary necessary for specific low-frequency isolation.

Question 4 – Comparison Between Three Porous Materials

The same computation is repeated for three *new* porous materials:

Material	ϕ	σ [$\text{N}\cdot\text{s}\cdot\text{m}^{-4}$]	α_h	Λ [μm]
Glass wool	0.97	25000	1.10	180
Rock wool	0.95	35000	1.15	150
Polyurethane foam	0.90	8000	1.20	250

Results. Figures 2, 3 and 4 show the real and imaginary parts of $\alpha(\omega)$ for the three sources and the different materials.

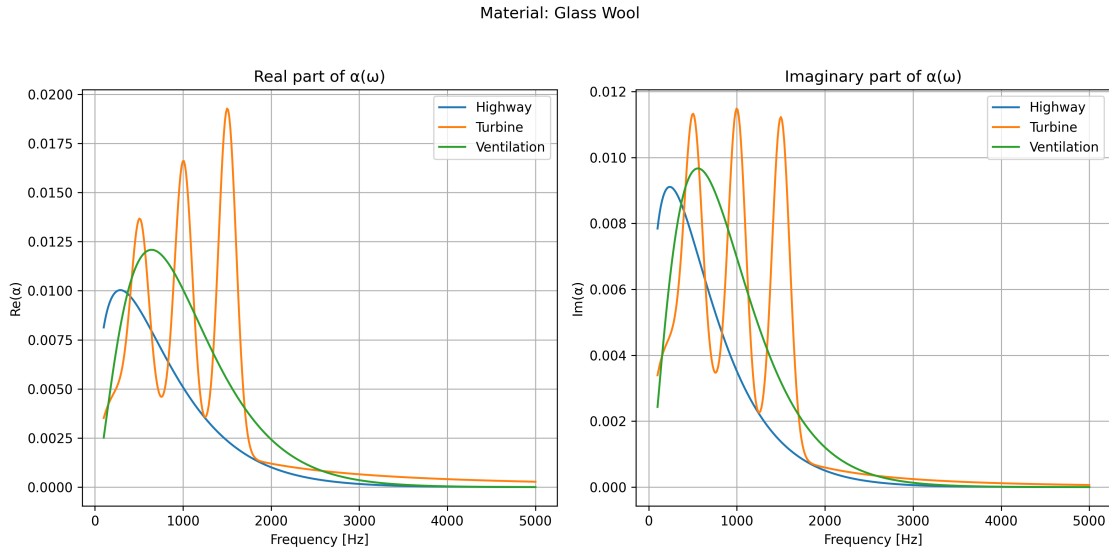


Figure 2: Glass wool – $\Re(\alpha)$ and $\Im(\alpha)$ vs. frequency for the three sources.

Material: Rock Wool

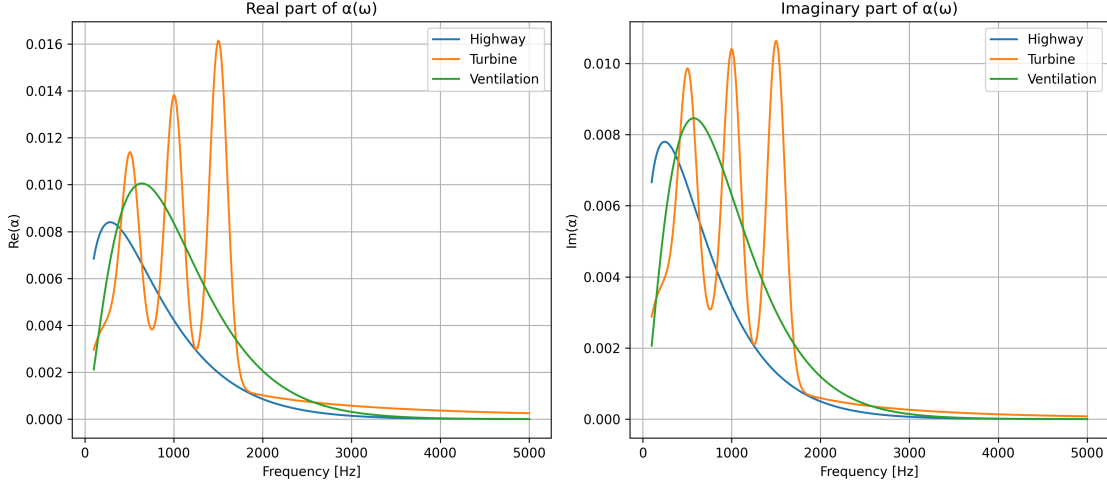


Figure 3: Rock wool – $\Re(\alpha)$ and $\Im(\alpha)$ vs. frequency for the three sources.

Material: Polyurethane Foam

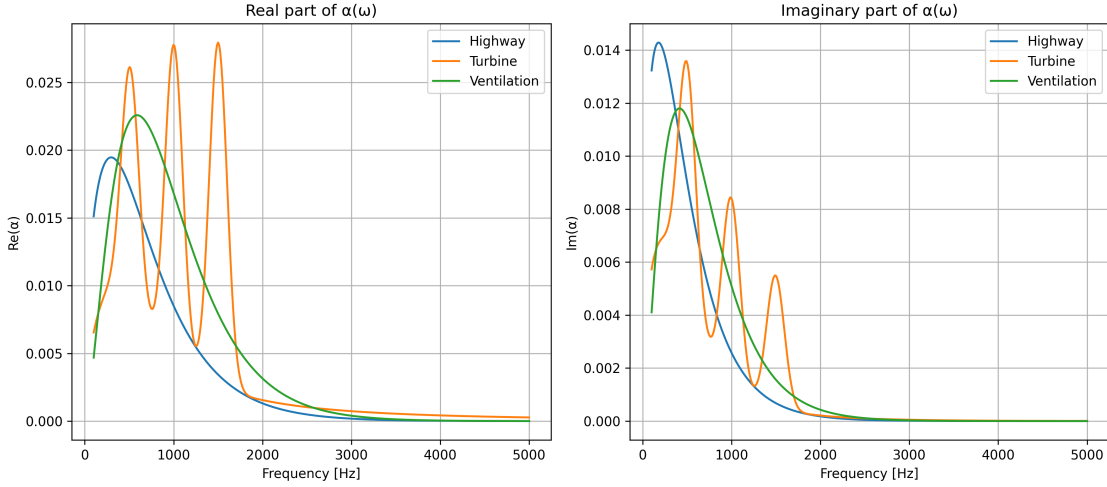


Figure 4: Polyurethane foam – $\Re(\alpha)$ and $\Im(\alpha)$ vs. frequency for the three sources.

Analysis. The optimal $\alpha(\omega)$ is inversely related to the porous material's surface impedance $Z_s(\omega)$. Material properties dictate the resulting $\Re(\alpha)$ (absorption efficiency) and $\Im(\alpha)$ (reactance).

- **Rock wool** ($\sigma = 35000$) exhibits very low $\Re(\alpha)$ across the spectrum, suggesting relatively low absorption compared to the other materials in this configuration.
- **Glass wool** ($\sigma = 25000$) provides a slightly better $\Re(\alpha)$ response.
- **Polyurethane foam** ($\sigma = 8000$) has the highest resistivity, leading to a significantly higher and broader $\Re(\alpha)$ profile. This material acts as a less reactive, more dissipative boundary over the tested frequency range, resulting in a superior overall absorption capacity.

The results confirm that the α coefficient successfully adapts to reproduce the distinct physical and acoustic characteristics of each porous material, allowing for an efficiency comparison based on the source spectrum.